

Real-World Clutter Improves Egocentric Distance Judgments in Optical See-Through Augmented Reality

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Abstract

Accurate distance perception is critical for effective interaction in augmented reality (AR), yet users often underestimate distances to virtual objects in optical see-through AR (OST-AR). While prior work suggests that environmental clutter can improve distance judgments in VR, it remains unclear whether clutter provides the same benefit in OST-AR, especially when the clutter is virtually rendered rather than physically present. To examine this question, we conducted a blind-walking study in which 36 participants, wearing a Microsoft HoloLens 2, judged egocentric distances across three between-subjects environments: No Clutter, Real-World Clutter, and AR Clutter. Real-World Clutter significantly reduced distance underestimation and produced near-veridical judgments, whereas AR Clutter did not significantly improve accuracy compared with the other environments. These findings suggest that physical objects in the surrounding environment can provide useful spatial references for judging virtual target distances in OST-AR, whereas virtual clutter may offer less reliable perceptual support. More broadly, this work highlights the importance of considering real-world environmental context when designing AR experiences that require accurate spatial perception and interaction.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; *Empirical studies in HCI*; User studies.

Keywords

Augmented Reality (AR), Egocentric Distance Perception, Optical See-Through AR, Environmental Clutter, Spatial Perception

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1 Introduction

Accurate distance perception is essential in extended reality (XR) applications such as training, simulation, and navigation, where users must act based on precise spatial information [Cooper 2023]. However, people often underestimate distances in ‘action space,’ (i.e., 2–30 m from the observer) in both virtual reality (VR) [Creem-Regehr et al. 2023; Cutting and Vishton 1995] and augmented reality (AR) [Adams et al. 2022, 2021; Bodenheimer et al. 2022; Rosales et al. 2019]. This underestimation can be influenced by display characteristics such as head-mounted display (HMD) weight, field of view (FOV), and pixel density [Buck et al. 2021, 2018; Creem-Regehr et al. 2023; Kelly 2023; Renner et al. 2013], as well as by the estimation method used, such as verbal estimation or blind walking [Creem-Regehr et al. 2023].

Beyond these general display and methodological factors, AR introduces additional challenges. Many OST-AR displays present virtual imagery at a fixed focal distance even when objects are rendered at different simulated depths, creating a conflict between where the eyes focus (accommodation) and converge (vergence) that may affect depth perception [Zhou et al. 2021]. For example, studies commonly report distance underestimation in optical see-through AR (OST-AR), where virtual objects may lack or conflict with real-world depth cues [Adams et al. 2022, 2021; Bodenheimer et al. 2022; Rosales et al. 2019]. Findings for video see-through AR (VST-AR), including handheld and head-mounted devices, are more varied, indicating that device and viewing configurations may affect distance judgments differently [Adams et al. 2022; Chakraborty et al. 2021b; Liu et al. 2020; Livingston et al. 2009; Swan et al. 2017]. More broadly, calibration, spatial registration, and inconsistencies



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between real and virtual depth cues, such as occlusion, may influence perceived distance in AR [Grubert et al. 2018; Itoh et al. 2021; Krajancich et al. 2020; Kruijff et al. 2010]. Although these technical factors are well documented, less is known about how the visual structure of the surrounding environment affects distance judgments.

One important aspect of environmental structure is clutter, which refers to objects surrounding a target and is common in everyday environments such as rooms and offices [Masnadi et al. 2023]. Nevertheless, many AR and VR distance-perception studies have used empty hallways, sparse rooms, or largely cueless virtual environments [Buck et al. 2021, 2018; Gagnon et al. 2021b; Kelly et al. 2017a; Rosales et al. 2019]. Existing evidence suggests that clutter could either hinder or support distance judgments. In AR, clutter can increase visual search time and draw attention away from relevant information [Flittner and Gabbard 2021; Kim et al. 2025]. In VR, however, environmental clutter can reduce distance underestimation [Masnadi et al. 2023]. Other visual structures, such as peripheral head-centered reference frames, can similarly improve distance-judgment accuracy by providing stable visual anchors, reducing sensory conflict, and supporting gaze stability [Hmaiti 2023; Hmaiti et al. 2023; Prothero 1998; Prothero and Parker 2003; Shi et al. 2021].

It therefore remains unclear whether clutter in AR improves distance judgments by providing useful spatial references or disrupts them by increasing visual complexity. It is also unclear whether physical and virtual clutter provide equally useful spatial information. Unlike physical objects, virtual objects may be affected by rendering, registration, and occlusion limitations, potentially making them weaker or less reliable spatial cues [Grubert et al. 2018; Krajancich et al. 2020; Kruijff et al. 2010]. To address these questions, we investigated *how virtual clutter affects distance judgments in OST-AR compared with real-world clutter*. Following Masnadi et al. [2023], we defined clutter as environmental objects surrounding the target. We compared distance judgments across three environmental conditions—no clutter, physical clutter, and virtual clutter—and across multiple target distances.

Our findings suggest that physical clutter supports more accurate distance judgments in OST-AR, whereas virtual clutter provides weaker or less reliable spatial information. These results indicate that distance perception in XR depends not only on display technology but also on the structure and source of the surrounding visual environment. For XR developers, this distinction highlights the potential value of using existing physical structures when placing virtual content to support more accurate spatial perception and interaction.

2 Related Work

2.1 Distance Perception in OST-AR

In OST-AR, users view the real world directly while virtual objects are overlaid on the display, altering how depth cues such as occlusion are presented [Grubert et al. 2018; Itoh et al. 2021; Krajancich et al. 2020; Kruijff et al. 2010]. Because of hardware limitations, OST-AR systems typically do not render shadows for virtual objects, removing an important depth cue [Adams et al. 2021]. OST-HMDs such as the Microsoft HoloLens 2 also have a restricted field of view,

which can constrain the availability of peripheral cues and affect spatial perception [Ren et al. 2016]. As a result, distance judgments in OST-AR can differ from those in the real world and in VST-AR displays, including handheld AR displays [Adams et al. 2022; Bodenheimer et al. 2022; Chakraborty et al. 2021b; Gagnon et al. 2021b; Liu et al. 2020; Livingston et al. 2009; Stefanucci et al. 2021; Swan et al. 2017; Swan II et al. 2007]. Prior work shows that lighting alignment and cast shadows can improve perceived ground contact for virtual objects and increase distance judgments [Adams et al. 2021; Bodenheimer et al. 2023; Gao et al. 2020]. Other appearance-based cues also matter: higher color contrast can improve depth-judgment accuracy, whereas texture density appears to have little effect [Liu et al. 2024]; object motion, color, and luminance can alter perceived depth, with brighter virtual colors often appearing closer [Ashtiani et al. 2023]; and higher luminance contrast between the real scene and virtual object can reduce depth-perception accuracy [Wang et al. 2024]. Although these studies demonstrate that object appearance and display-related cues influence distance perception in OST-AR, less is known about the role of the visual environment surrounding a virtual target. To address this gap, the current study examines surrounding environmental objects, or clutter, as a potential source of spatial information.

2.2 Clutter and Distance Perception

Everyday spaces are rarely empty; rooms, offices, and hallways typically contain furniture and other objects that provide a rich visual context [Masnadi et al. 2023]. This visual complexity may matter for AR because clutter can affect attention and task performance [Flittner and Gabbard 2021; Hmaiti et al. 2024; Kim et al. 2025]. For instance, increasing virtual clutter in AR can increase visual search time, indicating reduced search efficiency [Flittner and Gabbard 2021]. Kim et al. [2022] also found that physical objects were detected more readily than perceptually matched virtual objects during an AR search task, suggesting that users may process physical and virtual content differently even when they are designed to look similar. Thus, existing AR research indicates that clutter can increase attentional demands and that physical and virtual objects may not provide equivalent perceptual information. However, these studies do not establish how either form of clutter affects distance judgments.

In contrast, VR research suggests that surrounding visual structure can improve distance perception [Hmaiti 2023; Hmaiti et al. 2023; Masnadi et al. 2023, 2022]. In the study most directly related to ours, Masnadi et al. [2023] placed virtual furniture and room structures around targets and compared uncluttered, semi-cluttered, and cluttered versions of the same indoor and outdoor VR environments. Distance underestimation decreased as environmental clutter increased, and this improvement occurred across simulated FOVs ranging from 35° to 165°. This result suggests that the surrounding objects, rather than the visible extent of the display, supported more accurate distance judgments. Other studies have shown similar improvements using peripheral, head-centric reference frames, such as mesh masks, virtual noses, and hat brims [Hmaiti 2023; Hmaiti et al. 2023]. These head-centric reference frames reduced distance underestimation across different HMD FOVs, suggesting that stable peripheral visual references can support more accurate distance

judgments. One possible explanation is that surrounding visual information provides stable reference points that help users judge distances more accurately [Hmaiti 2023]. However, these findings are based entirely on VR, where both the target and the surrounding visual references are rendered within the same virtual environment. In OST-AR, physical clutter provides natural occlusion, shadows, and ground-contact cues, whereas virtual clutter may be affected by rendering, registration, and occlusion limitations [Grubert et al. 2018; Krajancich et al. 2020; Kruijff et al. 2010]. It therefore remains unclear whether clutter in OST-AR improves distance judgments or increases visual-attention demands, and whether physical and virtual clutter provide equally useful spatial information. To address these gaps, the current study compares distance judgments under no-clutter, physical-clutter, and virtual-clutter conditions in OST-AR.

2.3 Measuring Distance Perception in Action Space

To measure distance perception in ‘action space’ [Cutting and Vish-ton 1995], researchers have used tasks such as verbal reports, visual matching, blind walking, blind throwing, distance bisection, and triangle completion [Creem-Regehr et al. 2023; Kunz et al. 2009b; Philbeck and Loomis 1997; Sahm et al. 2005a]. These tasks capture perceived distance in different ways. Verbal reports require participants to estimate a distance in familiar units [Chakraborty et al. 2024a; Creem-Regehr et al. 2023; Gagnon et al. 2021a]; visual matching asks them to adjust a comparison stimulus to match the target distance [Loomis et al. 1992]; blind throwing asks them to throw an object to the remembered target location [Sahm et al. 2005a]; bisection asks them to indicate the midpoint between themselves and a target [Bodenheimer et al. 2007]; and triangle completion requires spatial updating as participants estimate a direct return path after walking two sides of a triangle [Loomis et al. 1993]. In blind walking, participants view a target, close their eyes, and walk to its remembered location [Philbeck and Loomis 1997]. Prior work suggests that blind walking can provide more accurate distance estimates than other tasks because it measures perceived distance through locomotor behavior rather than through explicit verbal or perceptual reports, which may be more susceptible to response bias [Creem-Regehr et al. 2023; Kunz et al. 2009a]. Because blind walking is widely used in VR and AR depth-perception research and provides a behavioral measure that is less susceptible to explicit reporting biases [Bandali et al. 2026; Chakraborty et al. 2024b; Creem-Regehr et al. 2005; Kelly et al. 2017b; Renner et al. 2013; Swan II et al. 2007], we used it to examine how physical and virtual clutter affect distance judgments in the current study.

3 Method

In this study, we investigated whether surrounding environmental objects, or clutter, enhance egocentric distance judgments in OST-AR and whether this effect differs between real and virtual clutter. We used a mixed experimental design with Environment (No Clutter, Real-World Clutter, AR Clutter) as a between-subjects factor and Target Distance (4 m, 5.5 m, 7 m) as a within-subjects factor. Thirty-six participants were randomly assigned to one of

the three Environment groups, with 12 participants per group. Participants completed 13 blind-walking trials, in which they viewed a virtual target cone at one Target Distance, saw it disappear, and then walked to its estimated location without visual feedback [Philbeck and Loomis 1997; Sahm et al. 2005b]. Distance judgment accuracy was quantified using Proportional Distance Error, calculated as the ratio of walked distance to actual target distance. We also administered the Simulator Sickness Questionnaire (SSQ) before and after the session to assess whether the task induced cybersickness [Kennedy et al. 1993]. Because the session was brief (12–15 minutes), most participants’ scores did not change; therefore, SSQ results are not reported further. Further details of the study design are provided in the following sections.

3.1 Hypotheses

Prior work has shown that egocentric distance judgments depend on the availability of visual depth cues, and that adding environmental clutter can improve distance judgments in action space [Masnadi et al. 2023, 2022]. Prior AR research also suggests that real-world objects provide more reliable depth information because they are physically present and visually grounded in the environment. In contrast, virtual objects are constrained by display, calibration, and rendering limitations, which can reduce the effectiveness of the depth cues they provide [Bodenheimer et al. 2022; Kruijff et al. 2010; Livingston et al. 2009; Swan et al. 2015]. Therefore, we expected AR Clutter to improve distance judgments compared with No Clutter, but not to provide the same perceptual support as Real-World Clutter. Prior work has shown that absolute distance underestimation increases with target distance in both OST-AR [Swan et al. 2007] and handheld AR [Chakraborty et al. 2021a,b]. However, in VR, proportional distance error may increase with target distance or remain relatively stable [Bandali et al. 2026; Chakraborty et al. 2024a,b]. We define this measure in section 4 as the ratio of estimated to actual distance. Based on these findings, we hypothesized that: **H1**: Adding surrounding objects, either real or virtual, will improve distance judgments to virtual targets compared with No Clutter. **H2**: Real-World Clutter will yield more accurate distance judgments than AR Clutter. **H3**: AR Clutter will yield better distance judgments than No Clutter, but worse than Real-World Clutter. **H4**: Proportional distance error will increase as target distance increases.

3.2 Participants

We conducted an a priori power analysis in G*Power for a repeated-measures within-between interaction ANOVA (F test; $f = 0.25$, $\alpha = 0.05$, power = 0.80, assumed correlation among repeated measures = 0.50, non-sphericity correction $\epsilon = 1.00$); three Environment groups: No Clutter, Real-World Clutter, and AR Clutter; and distance judgments at three Target Distances: 4 m, 5.5 m, and 7 m. The analysis indicated a required sample size of 36 participants (12 participants per Environment group). This sample size is consistent with prior studies investigating egocentric distance perception in AR and VR environments, which have successfully utilized similar numbers of participants (approximately 8–16 participants per between-subjects group in mixed-designs studies) to detect significant effects in blind-walking tasks [Masnadi et al. 2023; Swan II et al. 2007]. We recruited 36 participants (19 male, 17 female; $M = 23.50$ yrs, $SD = 3.91$ yrs)



Figure 1: The figure provides an overview of the experimental setup and the three environments. Image (a) shows the starting guideline (green line) and directional arrow (red arrow) rendered in the HoloLens 2 for all environments; participants stood behind the guideline at the start of each trial to standardize position and walking direction. Image (b) shows the No Clutter environment, with only the virtual target cone and guideline. Image (c) shows the Real-World Clutter environment, where the cone was overlaid onto a real-world hallway containing real chairs, recycling bins, and cardboard boxes. Image (d) shows the AR Clutter environment, where the cone was surrounded by virtual replicas of the same objects in the same locations. In all environments, participants viewed the cone and then completed a blind-walking distance judgment.

from our institution. In the No Clutter group, there were 8 males ($M = 27.13$ yrs, $SD = 3.44$ yrs) and 4 females ($M = 25.50$ yrs, $SD = 4.43$ yrs). The Real-World Clutter group consisted of 4 males ($M = 21.00$ yrs, $SD = 2.45$ yrs) and 8 females ($M = 20.13$ yrs, $SD = 2.03$ yrs). The AR Clutter group consisted of 7 males ($M = 26.14$ yrs, $SD = 4.34$ yrs) and 5 females ($M = 22.20$ yrs, $SD = 2.59$ yrs). Participants were recruited through our institution's online psychology study sign-up system and personal contacts, with no prior knowledge of the experiment. They received 2 course credits (2 credits/hour) as compensation for participation.

3.3 Equipment

3.3.1 Hardware and Software. The experiment was conducted using a Microsoft HoloLens 2 head-mounted display (HMD). It is an optical see-through augmented reality device equipped with see-through waveguide displays, integrated head tracking, hand tracking, eye tracking, and spatial mapping capabilities. The device provides a resolution of 2048×1080 pixels per eye, a refresh rate of 60 Hz, and an approximate diagonal FOV of 52° . Head pose and user movement were tracked using the device's onboard sensors and inside-out tracking system, allowing accurate registration of virtual content in the physical environment. The study was developed using the Unity Game Engine (2022.3.45f1) and deployed to the HoloLens 2 using the Mixed Reality Toolkit (MRTK 3; core package 4.0.0-pre.1). Virtual clutter objects were modeled in Blender (3.6 LTS).

3.3.2 Virtual Environments. The experiment included three environments: No Clutter, Real-World Clutter, and AR Clutter (see Figure 1). In all environments, participants judged the egocentric distance to a virtual traffic cone presented at one of five distances from the participant across trials: 4 m, 4.75 m, 5.5 m, 6.25 m, or 7 m. A horizontal green guideline ($2.5 \text{ m} \times 0.01 \text{ m}$) was rendered on the ground to indicate the participant's starting position for each trial. Participants were required to stand behind this line before beginning the trial. A red arrow on the ground, positioned at the participant's location behind the guideline, indicated the direction in which the virtual cone would appear on that trial, allowing participants to orient themselves appropriately. The guideline and arrow

are shown in Figure 1(a). The study was conducted in a long hallway. Participants were instructed to face the far end of the hallway, and both the virtual cone and the red arrow appeared in that direction. At the beginning of each trial, an invisible virtual ground plane was aligned with the real hallway floor at participants' feet. This plane served as a consistent reference for positioning the virtual cone and, in the AR Clutter environment, the virtual clutter objects at floor level. No virtual floor surface was rendered or visible to participants.

In the No Clutter environment (Figure 1(b)), the scene contained only the virtual traffic cone, the virtual ground plane, and the starting guideline with the directional arrow, overlaid on the physical hallway. In the Real-World Clutter environment (Figure 1(c)), participants saw the same virtual elements as in the No Clutter environment, but the physical hallway also contained several real-world objects, including different types of chairs, cardboard boxes, and trash bins. In total, the environment included approximately 25 office chairs (including rolling office chairs, stationary chairs, and wooden chairs), 11 cardboard boxes (both open and closed), and 12 trash bins (blue recycling bins). These objects were placed along the hallway walls, while the virtual cone was presented in the middle of the cluttered space at the five distances mentioned earlier. The minimum and maximum distances of the clutter objects from the participant's starting position were approximately 1.0 m and 12.5 m, respectively. The objects were randomly distributed on both sides of the hallway, with side-by-side spacing ranging from approximately 0.5 m to 1 m. In the AR Clutter environment (Figure 1(d)), participants saw the same virtual elements as in the No Clutter environment, along with virtual counterparts of the Real-World Clutter objects. These virtual objects were placed at the same spatial locations as the physical clutter to match the Real-World Clutter environment. In both the Real-World Clutter and AR Clutter environments, the clutter objects remained static throughout the experiment.

To ensure that the nearest AR Clutter objects were clearly visible within the limited field of view of the HoloLens 2, we maintained the minimum distance of 1.0 m between the participant's starting position and the nearest clutter object mentioned earlier. To preserve symmetry across environments, the Real-World Clutter

objects were also placed starting at this same minimum distance from the participant's starting position.

3.4 Procedure

Upon arrival, participants provided informed consent under a protocol approved by our Institutional Review Board and were randomly assigned to one of the three Environment groups. They first completed the pre-exposure SSQ to assess cybersickness before beginning the study and were then escorted by the experimenter to the hallway where the experiment was conducted. The hallway was about 100 m long and 3 m wide. We used a 13 m section of the hallway for the experiment. This provided enough space for blind walking up to the maximum target distance of 7 m. In the Real-World Clutter and AR Clutter conditions, objects were placed along both walls. This arrangement left a clear central walking path about 1.5 m wide.

A physical starting location was marked on the floor to correspond to the virtual starting position in the application. This location placed participants midway between the clutter objects on either side. The virtual cone always appeared straight ahead along the centerline, maintaining equal lateral spacing from the objects on both sides. Unlike previous studies that varied the cone's lateral position [Chakraborty et al. 2024a, 2021a,b], we kept it on the centerline to reduce the risk of collisions during blind walking. Once participants stood at the starting location and faced the far end of the hallway, they donned the headset and completed the headset's default five-point eye-calibration procedure, which calibrated the display to each user's inter-pupillary distance (IPD). We did not examine IPD as a factor because prior work found little to no effect on distance judgments in action space, which includes the distances tested in the current study [Chakraborty et al. 2024a]. The experimenter then helped the participants to launch the application on the headset and guided them through the study.

The application was controlled using three voice commands: 'Advance', 'Ready', and 'Here'. At the beginning of each trial, participants stood at the start location, faced the far end of the hallway, and said 'Advance', which initiated the trial. A virtual traffic cone (width = 0.64 m, height = 0.83 m, depth = 0.64 m) then appeared at one of the Target Distances described earlier. The size of the traffic cone remained constant across all trials. Participants were allowed as much time as needed to view the cone and, in the clutter environments, the surrounding environmental objects. When ready to make a distance judgment, they closed their eyes and said, 'Ready.' This command caused the cone to disappear in all environments, after which participants walked forward to the remembered cone location as accurately as possible, performing a blind-walking distance judgment task (see Figure 3(a)). Note that the virtual target cone disappeared in all environments, but the virtual clutter remained visible in the AR Clutter environment. Because the real-world clutter could not be removed during blind walking, we kept the virtual clutter visible in the AR Clutter environment to maintain a comparable experience across cluttered environments.

Because of the real-world clutter condition, an experimenter shadowed participants in all conditions for safety, consistent with prior work [Adams et al. 2018; Bandali et al. 2026; Kunz et al. 2013; Loomis et al. 1998]. When participants believed they had reached

the target location, they said 'Here' and stopped walking. The application automatically recorded the walked distance. However, because head movements at the start or stopping position could introduce slight noise into this measurement, we also obtained a physical measure of walked distance. A tape measure was placed along the side of the hallway, aligned with the physical start location. After the participant stopped, the experimenter measured the walked distance from the physical start location to the stopping point by placing a steel rod in front of the participant's shoes so that it perpendicularly intersected the tape measure, yielding the physical walked distance (see Figure 3(b)). Participants kept their eyes closed throughout the measurement and were then guided back to the start position; they opened their eyes only after returning to the start, ensuring they never observed the tape measure or their walked distance and received no feedback about their performance. We used this physical measure in our analysis.

After each trial, the experimenter guided participants back to the start location and reoriented them toward the far end, all with their eyes closed. Audio instructions delivered through the headset informed participants when they could open their eyes. At that point, the virtual green guideline and red directional arrow reappeared, indicating that the next trial could begin. Participants then verified that they were standing behind the guideline and facing the intended direction before repeating the same procedure. This process continued until all 13 trials were completed. The first two trials were practice trials, with the virtual cone presented at 4.75 m and 6.25 m to familiarize participants with the blind-walking procedure. The remaining 11 trials consisted of 9 actual trials (three each at 4 m, 5.5 m, and 7 m) and 2 dummy trials (one each at 4.75 m and 6.25 m). Trials were presented in a pseudo-randomized order such that (i) no two trials at the same distance occurred consecutively and (ii) no dummy trial appeared at the end. The dummy trials made the repeated experimental distances less predictable, thereby reducing potential learning effects. Practice and dummy trials were excluded from the analysis.

After completing all trials, participants again completed the SSQ to assess post-exposure cybersickness. The experimenter then thanked them and provided compensation. Figure 2 provides visual overviews of the overall experimental procedure. Participants completed the study in about 10 to 15 minutes on average.

4 Results

The primary outcome measure was Proportional Distance Error (PDE), calculated as the ratio of walked distance to actual target distance from the starting position. A PDE of 1.0 indicates accurate/veridical distance judgment, whereas values greater than 1.0 indicate overestimation and values less than 1.0 indicate underestimation. PDE was computed for each trial and then averaged across trials corresponding to each Target Distance for each participant, which yielded three observations per participant. Given 12 participants in each Environment group, this resulted in 36 observations per group and 108 observations in total. Before conducting inferential analyses, we screened for outliers, defined as observations lying more than ± 3 standard deviations from the grand mean. No outliers were identified, and all observations were retained. We

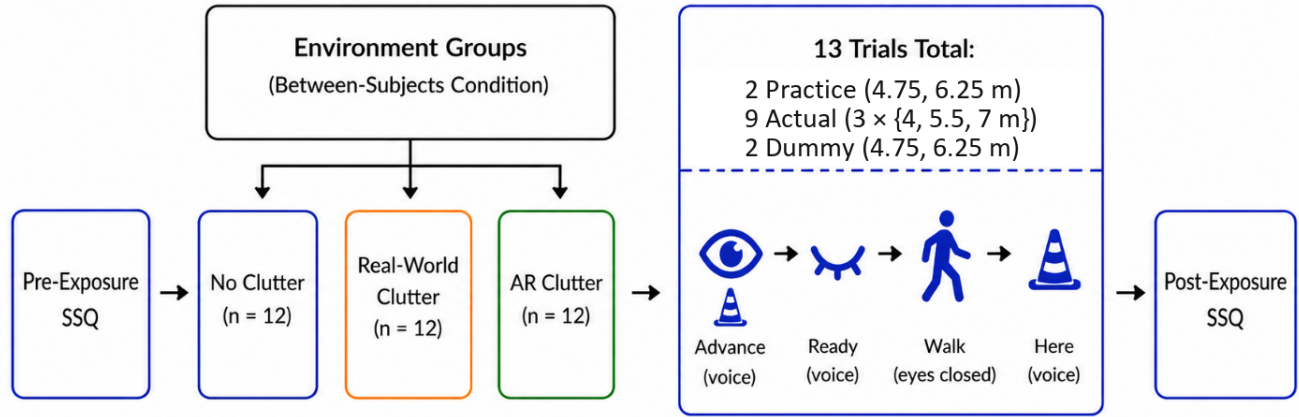


Figure 2: Outline of overall experimental procedure. Participants provided consent, completed the pre-exposure SSQ, and were randomly assigned to one of three Environment groups: No Clutter, Real-World Clutter, or AR Clutter. Participants then completed 13 blind-walking trials: 2 practice, 9 actual, and 2 dummy. In each trial, they viewed a virtual target and then walked with eyes closed to its remembered location. After all trials, participants completed the post-exposure SSQ.

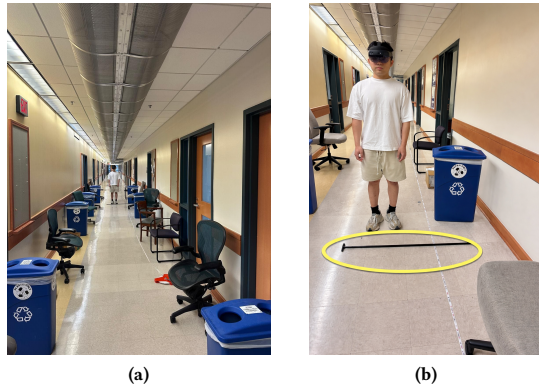


Figure 3: This figure shows the physical hallway setup. Image (a) shows the Real-World Clutter environment with chairs, recycling bins, and cardboard boxes. Image (b) shows a participant completing the blind-walking task with the HoloLens 2. Walked distance was measured using a floor-aligned tape measure, with a steel rod placed perpendicular to the tape at the participant's stopping point, as highlighted in the image.

then assessed normality using the Shapiro–Wilk test and homogeneity of variance using Levene's test. When these assumptions were violated, we applied the aligned rank transformation (ART) to the data [Wobbrock et al. 2011]. Analysis of variance (ANOVA) was then conducted in R on either the transformed or untransformed data, as appropriate. For analyses based on transformed data, we report descriptive statistics from the original, untransformed data to facilitate interpretation.

To examine accuracy of distance estimates, we conducted a 3 (Environment: No Clutter, Real-World Clutter, AR Clutter) $\times$ 3 (Target Distance: 4 m, 5.5 m, 7 m) mixed-design ANOVA on mean PDE. Environment was a between-subjects factor, and Target Distance was a within-subjects factor. Model residuals did not significantly deviate

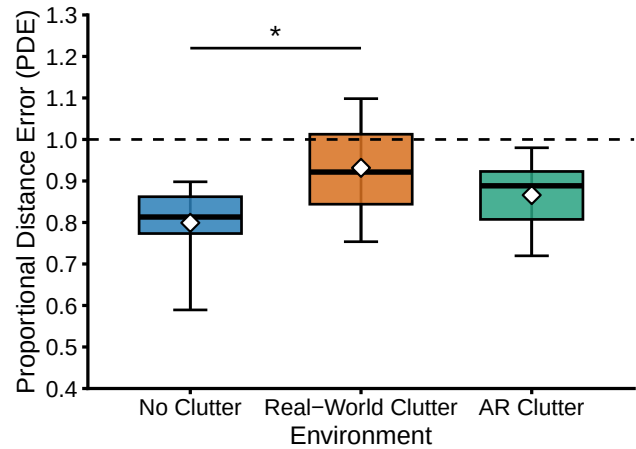


Figure 4: Distribution of Proportional Distance Error (PDE) across the three Environment groups. Boxes show the interquartile range, horizontal black lines inside boxes show medians, and white diamonds show means. Whiskers extend to the minimum and maximum values. The horizontal significance bar with an asterisk indicates a significant difference of $p < 0.05$, and the dashed horizontal line marks veridical performance (PDE = 1.0).

from normality ($W = 0.986$, $p = 0.324$), and Levene's test confirmed homogeneity of variance ($p = 0.478$). Therefore, we proceeded with the parametric mixed-design ANOVA.

The analysis revealed a significant main effect of Environment ($F(2, 30) = 4.21$, $p = 0.025$, $\eta_p^2 = 0.22$), indicating that distance judgment accuracy differed across Environment groups (see Figure 4). Pairwise comparisons with Holm correction showed that the participants in the No Clutter group ($M = 0.799$, $SD = 0.099$, $SE = 0.017$) had significantly lower PDE than the Real-World Clutter group (M

$= 0.932$, $SD = 0.124$, $SE = 0.021$; $t(30) = -2.89$, $p_{adj} = 0.021$). In contrast, the PDE of the participants in the AR Clutter group ($M = 0.866$, $SD = 0.090$, $SE = 0.015$) did not differ significantly from either the No Clutter group ($t(30) = -1.65$, $p_{adj} = 0.219$) or the Real-World Clutter group ($t(30) = 1.30$, $p_{adj} = 0.219$). There was also no significant main effect of Target Distance ($F(1.87, 55.55) = 0.83$, $p = 0.436$, $\eta_p^2 = 0.03$). Descriptive statistics of the Target Distance are as follows - at 4 m: $M = 0.860$, $SD = 0.123$, $SE = 0.021$; at 5.5 m: $M = 0.865$, $SD = 0.110$, $SE = 0.018$; and at 7 m: $M = 0.872$, $SD = 0.123$, $SE = 0.021$. We also found no significant interaction between Environments and Target Distance on mean PDEs ($F(3.70, 55.55) = 1.27$, $p = 0.292$, $\eta_p^2 = 0.078$).

Because frequentist statistics cannot confirm the null hypothesis when results are non-significant, we conducted Bayesian ANOVAs following Rouder et al. [2012] to further examine differences in distance judgments between (i) the AR Clutter and No Clutter groups and (ii) the AR Clutter and Real-World Clutter groups. In this analysis, Bayes Factors (BF_{01}) indicate the relative evidence in favor of the null hypothesis over the alternative hypothesis. A BF_{01} of 1 indicates equal support for both hypotheses, whereas values greater than 3, 10, or 30 indicate increasing evidence for the null hypothesis. Because Bayes Factors are sensitive to prior odds [Liu and Aitkin 2008], we set the prior odds to 1, giving equal prior weight to both hypotheses [Buck et al. 2018]. The BF_{01} values were 0.39 for AR Clutter versus No Clutter and 0.64 for AR Clutter versus Real-World Clutter. These results align with the frequentist analyses, suggesting no clear evidence that distance judgments differed between AR Clutter and either comparison group.

Overall, these findings suggest that participants underestimated all target distances. However, the degree of underestimation was significantly smaller in the Real-World Clutter group than in the No Clutter group, indicating that real-world spatial cues supported more accurate distance judgments. In contrast, AR Clutter did not improve distance judgments relative to the No Clutter environment. This pattern suggests that the AR clutter may not have provided sufficiently effective spatial cues for distance judgments, possibly because it lacked certain real-world depth cues, such as shadows [Adams 2020; Adams et al. 2022, 2021; Rosales et al. 2019].

5 General Discussion

This study examined how surrounding environmental objects, i.e., real-world versus virtual (AR) clutter, affect egocentric distance perception in an OST-AR head-mounted display. Our findings partially supported **H1**. Participants underestimated distances (with the blind walking task) more in the No Clutter environment than in the Real-World Clutter environment, but not relative to AR Clutter. The results fully supported **H2**, showing greater underestimation in No Clutter compared to Real-World Clutter. However, **H3** was not supported, as distance judgments in AR Clutter did not differ significantly from either No Clutter or Real-World Clutter. **H4** was also not supported, as distance errors did not differ significantly across target distances. These findings refine our understanding of the role of environmental cues in AR perception. They suggest that the perceptual system does not treat virtual clutter as an equivalent substitute for physical structure, even when it is designed to closely resemble it.

Our results are consistent with prior VR work showing that clutter can improve distance judgments in action space [Hmaiti 2023; Hmaiti et al. 2024, 2023; Masnadi et al. 2023, 2022], but they also extend this line of research to AR by distinguishing between real and virtual clutter. Our findings suggest that only real-world clutter in AR provides this benefit. One possible explanation is that virtual objects in OST-AR lack important depth cues, such as cast shadows, which provide ground contact perception that helps to judge distances accurately [Adams 2020; Adams et al. 2022, 2021; Bodenheimer et al. 2023; Rosales et al. 2019]. Additionally, limitations in occlusion and spatial registration may reduce the reliability of virtual clutter as a reference frame [Grubert et al. 2018; Krajancich et al. 2020; Kruijff et al. 2010]. Because the target and AR clutter were rendered through the same OST display, both were subject to the same luminance and transparency artifacts and accommodation conflicts, potentially limiting the clutter's usefulness as an independent spatial reference [Itoh et al. 2021; Kruijff et al. 2010]. In contrast, real-world clutter preserves these depth cues and provides stable spatial references, which likely helped participants estimate distances more accurately.

This interpretation also aligns with prior AR work showing that users may process real and virtual content differently. Kim et al. [2022] found that physical objects were detected more readily than visually matched virtual objects during AR search tasks. Similarly, in a different study, Kim et al. [2025] showed that increasing virtual content can reduce awareness of salient physical objects, suggesting that physical and virtual elements may not contribute equally to users' perception of the scene. Our findings extend this idea to distance perception: even though the AR Clutter was geometrically matched to the Real-World Clutter environment, it did not provide the same benefit for distance judgments. This suggests that virtual clutter may not provide depth information that is as reliable or spatially useful as real-world clutter, making it less effective as a reference for distance scaling in OST-AR. This difference may also explain why clutter could be more useful for judging virtual targets in OST-AR than physical targets in the real world. Physical targets are naturally integrated with the ground and surrounding environment and can be judged accurately when real-world depth cues are available [Philbeck and Loomis 1997]. In contrast, virtual targets may contain incomplete or conflicting depth cues and may therefore benefit more from reliable physical references [Kruijff et al. 2010; Stefanucci et al. 2021]. However, this explanation remains tentative because our study did not directly compare physical and virtual targets.

An experimenter shadowed participants in all three environments for safety. While this shadowing could in principle have influenced perception-action judgments, it is a standard practice in many distance-judgment experiments.

Another possible explanation is that the physical objects in the Real-World Clutter environment improved the HoloLens 2's inside-out tracking. Because the HoloLens uses visible environmental features to estimate its position, better tracking could have kept the virtual cone more stably registered with the physical hallway. However, tracking differences are unlikely to fully explain our results. The approximately 0.13 PDE difference between the No Clutter and Real-World Clutter environments corresponds to about 0.7 m at 5.5 m, which is substantially larger than the approximately 2

cm spatial accuracy reported for the HoloLens 2 [Soares et al. 2021]. Similar benefits of clutter have also been observed in VR, where this OST-AR tracking issue does not apply [Hmaiti 2023; Masnadi et al. 2023]. Nevertheless, because we manually aligned the virtual start marker with the physical starting location, small differences in registration stability across environments cannot be completely excluded. Future studies could use an external tracking system to directly measure the displayed target position and quantify tracking drift across environments.

Although the HoloLens 2 has a relatively small FOV of 52°, prior VR work has shown that clutter can improve distance judgments across a broad range of FOVs (35°–165°) [Masnadi et al. 2023]. Therefore, we do not believe FOV alone explains why AR Clutter failed to improve distance judgments in our study; however, future work should directly examine how FOV interacts with clutter in OST-AR. The weight of the HMD could also contribute to distance underestimation, but this effect would likely influence all environmental clutter groups rather than explain differences between them [Buck et al. 2021, 2018]. Finally, in both real and virtual clutter environments, the clutter objects were identical and placed in the same locations throughout the study. In the AR Clutter environment, virtual clutter remained visible after the virtual cone disappeared because real-world clutter also remained present during blind walking. This helped maintain comparable experiences across environments and minimize procedural confounds that could affect distance judgments across Environment groups.

Participants underestimated distances by approximately 25.16% in the No Clutter environment, 7.30% in the Real-World Clutter environment, and 15.47% in the AR Clutter environment. Across all environments, the mean underestimation rate with the HoloLens 2 was approximately 15.98%, which is lower than the mean underestimation rate of 26.52% reported across several VR HMDs by Kelly [2023]. This suggests that users may underestimate distances less in OST-AR than in VR, possibly because they can still perceive real-world depth cues. Adding real-world clutter further improved distance judgments, allowing participants to estimate virtual target distances significantly more accurately. Our overall underestimation rate is also consistent with prior HoloLens 2 findings: Adams et al. [2022] reported underestimation rates of ~16.7% with virtual object shadows and ~18.6% without shadows, while Swan II et al. [2007] reported ~14% underestimation with blind walking and ~23% with verbal estimation. Similarly, Masnadi et al. [2023] found that underestimation decreased slightly from ~23.4% in an uncluttered VR environment to ~22.7% in a semi-cluttered environment and ~22.4% in a cluttered environment. Consistent with Masnadi et al.'s finding that environmental clutter can improve distance judgments in VR, our results suggest that real-world clutter can also support more accurate distance judgments in OST-AR.

6 Limitations and Future work

Our between-subjects design included 12 participants per Environment group. Although this sample size was supported by an a priori power analysis for a medium effect and is comparable to prior studies using 8–16 participants per group [Masnadi et al. 2023; Swan II et al. 2007], it may have limited our ability to detect smaller

differences, particularly between AR Clutter and the other environments. Future studies with larger samples or within-subjects designs could determine whether AR Clutter provides a small but reliable benefit. Our findings are also limited to the HoloLens 2 and may not generalize to other AR devices. VST-AR HMDs can produce greater distance underestimation than OST-AR [Adams et al. 2022], whereas handheld AR studies have reported relatively accurate judgments [Chakraborty et al. 2021a; Liu et al. 2024; Swan II 2021]. Future work should therefore examine the effects of clutter across AR platforms. In addition, our study used a fixed arrangement of static clutter in a relatively narrow indoor hallway. Because distance judgments can vary with environmental scale and context [Buck et al. 2018; Kelly et al. 2022, 2017b; LaViola et al. 2017], the real-clutter advantage should be tested in differently sized indoor and outdoor spaces. Future research should also examine dynamic clutter, such as pedestrians and vehicles. Although prior handheld AR studies found that moving avatars did not affect distance judgments [Chakraborty et al. 2021a,b], it remains unclear whether moving environmental objects affect distance perception in OST-AR or VST-AR HMDs.

7 Conclusions

This study investigated whether physical and virtual surrounding objects support egocentric distance perception in OST-AR. Using a blind-walking task, we found that real-world clutter reduced distance underestimation and led to near-veridical judgments compared with no clutter, whereas AR clutter did not significantly improve performance. These findings suggest that physical clutter provides more reliable spatial cues for judging virtual object distances than virtual clutter, which highlights the importance of leveraging real-world context when placing AR content if accurate distance perception is needed.

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